

Logistic regression model

Evert Jan Karman

January 1900

Logistic regression model

1 Model

We obtain n data points with d features.

We convert the output to values -1 and 1. That will be stored in $n \times 1$ matrix Y .

We may remove the reviews labeled neutral from our dataset (the data with known sentiment)

In module 2 d features means how often d -th word appears in the original review text

We insert a dummy column with ones as 0-th column which we'll map to the intercept w_0

So we put the features in $n \times d$ matrix H

We also have $d + 1$ coefficients which we put in a $(d+1) \times 1$ matrix w where w_0 is the intercept

The scores per data point is obtained by multiplying H with w :

$Hw = \text{score}$ will be a $n \times 1$ matrix

After that we convert each single score to a probability using sigmoid function:

$$P = \frac{1}{1 + e^{-\text{score}}}$$